FUSIDIC ACID

Topical max 10 days

Never with statins (systemic)

CHEMICAL NAME (2Z)-2-[(3R,4S,5S,8S,9S,10S,11R,13R,14S,16S)-16-acetyloxy-3,11-dihydroxy-4,8,10,14-tetramethyl-2,3,4,5,6,7,9,11,12,13,15,16-dodecahydro-1H-cyclopenta[a]phenanthren-17-ylidene]-6-methylhept-5-enoic acid (sodium fusidate)

CLASS Fusidane (steroid-like) antibiotic

TRADE NAMES Fucidin · Fucithalmic (eye drops) · Fucicort (with betamethasone)

THERAPEUTIC USES / INDICATIONS

- Topical: impetigo, infected eczema, folliculitis, minor staphylococcal skin infections.
- Eye drops: bacterial conjunctivitis.
- Systemic: staphylococcal osteomyelitis, MRSA (combined with another agent).

PHARMACODYNAMICS

- Inhibits **elongation factor G** → blocks bacterial protein synthesis (bacteriostatic).
- Active against *Staph. aureus* incl. MRSA; resistance develops quickly when used alone.

ADMINISTRATION

- **Topical** cream/ointment 3–4 times daily for max 7–10 days (resistance).
- **Eye drops** twice daily.
- **PO** tablets/suspension with food; **IV** infusion over 2 h into large vein (phlebitis).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Topical: local irritation, itching.
- Systemic: nausea, diarrhoea, dyspepsia.

ADVERSE EFFECTS (SERIOUS)

- Systemic: jaundice and hepatotoxicity; thrombophlebitis (IV).
- Rhabdomyolysis with statins (systemic).
- Hypersensitivity reactions.

DOSAGES

- **Topical:** apply 3–4 times daily for 7–10 days.
- **Eye:** 1 drop twice daily for 7 days.
- **Systemic (adult):** 500 mg sodium fusidate PO every 8 h (with rifampicin or another agent).
- **Children:** suspension 15 mg/kg every 8 h.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- Systemic use with statins (stop statin during and 7 days after).
- **Caution:** hepatic impairment, biliary disease (systemic).

KEY INTERACTIONS

- **Statins:** fatal rhabdomyolysis (systemic fusidic acid).
- **HIV protease inhibitors:** ↑ levels of both.

PHARMACOKINETICS

- **Peak:** PO 2–4 h. Good bone and tissue penetration.
- Metabolised in liver; excreted in bile.
- **t½:** 5–6 h.

NURSING MANAGEMENT

- Topical: clean area, apply thin layer; limit course duration.
- Systemic: check statin use; monitor LFTs.
- Assess lesions for healing.
- Infection-control measures for impetigo.

PATIENT EDUCATION

- Use the cream only for the prescribed days.
- Wash hands before and after applying.
- Impetigo: avoid sharing towels; children stay home until crusts dry or 48 h after starting treatment.
- Report yellow skin or muscle pain (tablets).

KEY POINTS

- Anti-staphylococcal; mainly topical.
- Systemic: hepatotoxic; never with statins.
- Short courses to limit resistance.
- Combine with another drug when systemic.